NUMBER PATTERNS

Short Revision Notes — Create Level (Produce Something New)

1. Creating a Sequence From Given Conditions

Condition	Example	Created Sequence
First term = 5, common difference = 3	Start at 5, add 3 each time	5, 8, 11, 14, 17, ...
First term = 20, common difference = -5	Start at 20, subtract 5 each time	20, 15, 10, 5, 0, -5, ...
First term = 7, common difference = 0	Start at 7, add 0 each time	7, 7, 7, 7, 7, ...
First term = 100, common difference = -10	Start at 100, subtract 10 each time	100, 90, 80, 70, 60, ...
First term = 3, common difference = 4	Start at 3, add 4 each time	3, 7, 11, 15, 19, ...

Method: Start with the first term · Add the common difference repeatedly · Write terms separated by commas · End with ellipsis (...)

2. Creating the General Term From Conditions

Given Information	Created General Term
First term = 5, d = 3	T_n = 5 + (n–1)3 = 3n + 2
First term = 10, d = 7	T_n = 10 + (n–1)7 = 7n + 3
First term = 22, d = -3	T_n = 22 + (n–1)(-3) = 25 – 3n
First term = 8, d = 6	T_n = 8 + (n–1)6 = 6n + 2
First term = 15, d = -4	T_n = 15 + (n–1)(-4) = 19 – 4n

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Tn = First term + (n - 1) x common difference
Tn = dn + (First term - d)
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3. Creating Your Own Number Patterns

Pattern Type	Example	General Term
Even numbers	2, 4, 6, 8, 10, ...	T_n = 2n
Odd numbers	1, 3, 5, 7, 9, ...	T_n = 2n – 1
Multiples of 3	3, 6, 9, 12, 15, ...	T_n = 3n
Multiples of 5	5, 10, 15, 20, 25, ...	T_n = 5n
Multiples of 7	7, 14, 21, 28, 35, ...	T_n = 7n
Powers of 2	2, 4, 8, 16, 32, ...	NOT arithmetic (no common difference)

Create Your Own:

Task	Created Sequence	General Term
Multiples of 4	4, 8, 12, 16, 20, ...	T_n = 4n
Multiples of 6	6, 12, 18, 24, 30, ...	T_n = 6n
3, 10, 17, 24, 31, ...	d = 7, first = 3	T_n = 7n – 4

4. Creating Word Problems

Template	Example Problem
Distance Pattern	A cyclist rides 10 km on day 1. Each day he increases his distance by 5 km. How far does he ride on day 8?
Money Pattern	Sam saves Rs 50 in week 1. He saves Rs 15 more each week. How much does he save in week 12?
Seating Pattern	A theatre has 20 seats in row 1. Each row has 4 more seats than the previous row. How many seats in row 15?
Matchstick Pattern	Figure 1 uses 4 matchsticks. Figure 2 uses 7 matchsticks. Figure 3 uses 10 matchsticks. How many sticks in figure 20?

5. Creating Complex Sequences

Design Task	Created Sequence	General Term
Start at 100, decrease by 15	100, 85, 70, 55, 40, ...	$T_n = 115 – 15n$
Start at -5, increase by 6	-5, 1, 7, 13, 19, ...	$T_n = 6n – 11$
Start at 50, decrease by 8	50, 42, 34, 26, 18, ...	$T_n = 58 – 8n$
Start at 200, decrease by 25	200, 175, 150, 125, 100, ...	$T_n = 225 – 25n$

6. Creating Your Own Matchstick Patterns

Pattern 1: Triangular Pattern

Figure	Sticks	Calculation
Figure 1	3	1+2 = 3
Figure 2	6	2+3+1 = 6
Figure 3	9	3+3+3 = 9
Figure 4	12	3+3+3+3 = 12
General Term	$T_n = 3n$	

Pattern 2: Square Pattern

Figure	Sticks	Calculation
Figure 1	4	4 × 1
Figure 2	8	4 × 2
Figure 3	12	4 × 3
Figure 4	16	4 × 4
General Term	$T_n = 4n$	

Pattern 3: Growing Square Pattern

Figure	Sticks	Calculation
Figure 1	6	2(1) + 4 = 6
Figure 2	10	2(3) + 4 = 10
Figure 3	14	2(5) + 4 = 14

Figure	Sticks	Calculation
General Term	$T_n = 4n + 2$	

7. Creating the (n+1)th Term and Other Derivatives

Given T_n	Create T_{n+1}	Create T_{n-1}	Create T_{2n}
$T_n = 3n + 2$	3n + 5	3n – 1	6n + 2
$T_n = 5n – 3$	5n + 2	5n – 8	10n – 3
$T_n = 7n + 1$	7n + 8	7n – 6	14n + 1
$T_n = 20 – 3n$	17 – 3n	23 – 3n	20 – 6n
$T_n = 4n – 1$	4n + 3	4n – 5	8n – 1

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Tn+1 = Tn + d
Tn-1 = Tn - d
T2n = substitute 2n for n
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8. Creating Sequences With Pattern Qualities

Design Goal	Created Sequence	Why It Works
All terms positive	$T_n = 3n + 2$ (5, 8, 11, 14...)	First term is positive, d > 0
All terms negative	$T_n = 1 – 4n$ (-3, -7, -11, -15...)	First term is negative, d < 0
Mixed positive/negative	$T_n = 20 – 5n$ (15, 10, 5, 0, -5...)	Crosses zero at n=5
Alternating signs	NOT possible with common difference	Need multiplication pattern
All terms even	$T_n = 2n$ (2, 4, 6, 8, 10...)	d = 2, even terms
All terms odd	$T_n = 2n – 1$ (1, 3, 5, 7, 9...)	d = 2, odd terms

9. Creating Real-Life Scenarios

Scenario 1: Savings Pattern

```
Problem: A student saves Rs 20 in week 1, Rs 30 in week 2, Rs 40 in week 3, and so on.

a) Write the sequence: 20, 30, 40, 50, 60, ...
b) Common difference = 10
c) General term: Tn = 10n + 10
d) Week 20: T20 = 10(20) + 10 = Rs 210
```

Scenario 2: Temperature Pattern

```
Problem: Temperature drops from 30°C by 2°C each hour.

a) Write the sequence: 30, 28, 26, 24, 22, ...
b) Common difference = -2
c) General term: Tn = 32 - 2n
d) 12th hour: T12 = 32 - 24 = 8°C
```

Scenario 3: Construction Pattern

```
Problem: A builder builds 5 houses in year 1, 8 in year 2, 11 in year 3...

a) Write the sequence: 5, 8, 11, 14, 17, ...
b) Common difference = 3
c) General term: Tn = 3n + 2
d) Year 10: T10 = 3(10) + 2 = 32 houses
```

10. Creating Your Own Problem Set

Level 1: Simple Sequence Creation

Task	Your Answer
Create a sequence starting at 5, common difference 4	5, 9, 13, 17, 21, ...
Create a sequence starting at 100, common difference -7	100, 93, 86, 79, 72, ...
Write the general term for a sequence with T_1 = 8 and d = 5	T_n = 5n + 3

Level 2: Term Finding Problems

Task	Your Answer
Find the 25th term of T_n = 4n – 1	T_{25} = 4(25) – 1 = 99
Which term of T_n = 6n + 4 equals 100?	6n+4=100 → n=16 (16th term)
Find the 50th term of T_n = 120 – 3n	T_{50} = 120 – 150 = -30

Level 3: Word Problem Creation

Task	Example
Create a salary pattern problem	A teacher earns Rs 45,000 in year 1. Salary increases by Rs 3,000 each year. Find salary in year 5.
Create a seating pattern problem	Cinema row 1 has 8 seats. Each row adds 3 seats. Find seats in row 20.
Create a distance pattern problem	A train travels 80 km in hour 1, increases by 15 km each hour. Find distance in hour 10.

11. Creating Complex General Terms

From First Term and Common Difference

First Term	Common Difference	General Term Created
5	3	$T_n = 3n + 2$
12	7	$T_n = 7n + 5$
25	-4	$T_n = 29 – 4n$
-3	6	$T_n = 6n – 9$
40	-10	$T_n = 50 – 10n$

From Two Known Terms

Known Terms	Find d	General Term
$T_1 = 5$, $T_3 = 11$	d = (11-5)÷2 = 3	$T_n = 3n + 2$
$T_2 = 13$, $T_5 = 25$	d = (25-13)÷3 = 4	$T_n = 4n + 5$
$T_1 = 20$, $T_4 = 8$	d = (8-20)÷3 = -4	$T_n = 24 – 4n$

12. Creating Sequences With Specific Properties

Property	Created Sequence	General Term
First 3 terms are 5, 9, 13	5, 9, 13, 17, 21...	$T_n = 4n + 1$
Common difference is 6	2, 8, 14, 20, 26...	$T_n = 6n – 4$
First term is 15, 20th term is 110	$T_{20} = 15 + 19d = 110 \rightarrow 19d = 95 \rightarrow d = 5$	$T_n = 5n + 10$
Terms decrease by 8 starting from 50	50, 42, 34, 26, 18...	$T_n = 58 – 8n$
Sum of first 2 terms is 15, d = 4	$T_1 + T_2 = a + (a+4) = 15 \rightarrow 2a=11 \rightarrow a=5.5$	$T_n = 4n + 1.5$

13. Creating Your Own Sequence Design

Design Template 1: Increasing Pattern

```
Pattern: Every term increases by _______

First term: _____
Common difference: _____
Sequence: _____________________
General term: Tn = _______
20th term: _______
```

Design Template 2: Decreasing Pattern

```
Pattern: Every term decreases by _______

First term: _____
Common difference: _____
Sequence: _____________________
General term: Tn = _______
15th term: _______
```

Design Template 3: Real-World Pattern

```
Context: ________________________________

Sequence: _____________________
Common difference: _____
General term: Tn = _______
Question to answer: _________
Answer: _________
```

14. Creative Extension: Design Your Own Sequence

Criteria	Your Design
Start	Choose a first term between 1 and 100
End	Choose a common difference between -10 and +10
Sequence	Write first 5 terms
General Term	Write T_n formula
Question	Ask: “Which term equals _______?”
Answer	Solve and justify

Example:

```
First term = 25
Common difference = -3
Sequence: 25, 22, 19, 16, 13, ...
General term: Tn = 28 - 3n
Question: Which term equals -5?
Solution: 28 - 3n = -5 -> 3n = 33 -> n = 11 (11th term)
Verification: T11 = 28 - 33 = -5 (correct)
```

15. Self-Test — Creation Questions

No.	Task	Your Answer
1	Create a sequence with $T_1 = 8$, $d = 5$	8, 13, 18, 23, 28, ...
2	Write T_n for sequence 10, 17, 24, 31...	$T_n = 7n + 3$
3	Create a word problem about seating	A hall has 12 seats in row 1, adds 4 each row. How many in row 15?
4	Find T_{n+1} if $T_n = 3n – 2$	$T_{n+1} = 3n + 1$
5	Create a decreasing sequence with d = -6	30, 24, 18, 12, 6, 0, ...
6	Write T_n for a sequence with $T_1 = 7$, $d = 4$	$T_n = 4n + 3$
7	Create a matchstick pattern with $T_n = 5n + 2$	Figure 1=7, Figure 2=12, Figure 3=17, Figure 4=22
8	Find T_{20} for $T_n = 10n – 5$	$T_{20} = 200 – 5 = 195$
9	Create a sequence with first 3 terms 8, 14, 20	8, 14, 20, 26, 32, ...
10	Write T_n for a sequence where T_1=100 and T_{10}=10	d = (10-100)÷9 = -10; $T_n = 110 – 10n$

16. Creation Checklist

When creating a sequence:

- Choose first term and common difference
- Add d repeatedly to get terms
- Check: $T_2 – T_1 = d$, $T_3 – T_2 = d$
- Write with commas and ellipsis (...)
- Write the general term

When creating a general term:

- Identify d (coefficient of n)
- Find c (constant term)
- Write $T_n = dn + c$
- Verify with n=1, n=2

When creating word problems:

- Choose a real-world context
- Identify first term and d
- Create a clear question
- Provide the solution

When creating derivatives:

- $T_{n+1} = T_n + d$
- $T_{n-1} = T_n – d$
- T_{2n} = substitute 2n
- $T_{n+k} = T_n + kd$

17. Creation Templates

Template 1: Create From First Term and d

```
First term (a) = _____
Common difference (d) = _____
Sequence: _____, _____, _____, _____, _____...
General term: Tn = _____
```

Template 2: Create From Two Terms

```
T1 = _____
T3 = _____
Find d = (T3 - T1) / 2 = _____
Sequence: _____, _____, _____, _____, _____...
General term: Tn = _____
```

Template 3: Create Word Problem

```
Context: ________________________________
First term: _____
Common difference: _____
Sequence: _____, _____, _____, _____, _____...
Question: ________________________________
Solution: ________________________________
Answer: ________________________________
```

Template 4: Create the (n+1)th Term

```
Given: Tn = _____
Tn+1 = _____
Tn-1 = _____
T2n = _____
```

Revision Tip: *Creation is the highest level of thinking! Be creative, test your own sequences, and always verify your work by substituting back. The best way to master patterns is to create them yourself!*